

Owen Koehler

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Education

Northeastern University, Boston, MA

Candidate for Bachelor of Science in Mechanical Engineering

May 2028

GPA: 3.9

Courses: Dynamics, Fluid Mechanics, Measurements and Analysis, Thermodynamics, Mechanics of Materials, Statics

Activities: Northeastern Electric Racing, ASME Northeastern, Rock Climbing, Basketball

Skills

Applications/Languages: SolidWorks (CSWA), AutoCAD, Fusion 360, C++ (Arduino), Proficient in MATLAB

Tools: CNC Router, 3D Printing (SLA and FDM), Laser Cutting, MIG Welding, Soldering

Engineering: FEA, Inverse Kinematics, Embedded Systems, Stepper Motor Control

Work Experience

Abbott Heart Failure, Pleasanton, CA

Jan 2026 - Aug 2026

Systems Engineering Co-Op

- Led a team of 4 co-ops to complete an automated LVAD power cycling fixture that had been stalled since 2023, eliminating manual operator intervention and saving an estimated \$750K annually
- Redesigned and upgraded the fixture's hardware, adding an SD-card data logging system and a rotary encoder with a push-button for user input allowing for modularity that was not previously possible
- Rewrote the fixture's control software from the ground up (Arduino, C++) into a modular architecture capable of dynamically handling 0–5 batteries based on real-time user input
- Engineered an autonomous data/debug logging system to microSD, structuring output for direct import into Excel to plot battery capacity degradation against cycle count for engineering analysis
- Performed equipment calibration and HeartMate 3 LVAD verification testing, plus outflow graft permeability testing to assess sealant performance and biocompatibility under physiological flow

Summit Bicycles, Palo Alto, CA

May 2021 - Mar 2024

Sales Associate

- Sold over \$100,000 in volume working part-time and ensured customer satisfaction with every sale
- Performed basic bike repairs and builds for both average consumers and professional bike teams

Projects and Activities

4-DOF Robotic Arm – Independent Project

Apr 2026 - Aug 2026

- Designed, 3D-printed, and built a 4-DOF robotic arm with 270 mm reach, ~500 g payload, and ~1 mm repeatability for under \$200, with a custom GUI for pick-and-place route programming and homing
- Designed cycloidal reducers for NEMA 17 steppers, minimizing backlash while keeping backdrivability
- Implemented inverse kinematics and modified GRBL firmware to integrate the gripper into stepper control

Northeastern Electric Racing R&D for Vehicle Dynamics Subteam

Sep 2025 - Present

- Researched and designed systems for mounting the steering rack and steering column on the FSAE vehicle
- Applied principles of vehicle dynamics and structural design to ensure performance, safety, and manufacturability
- Collaborated with a multidisciplinary team to integrate designs with suspension and chassis systems

Energy Museum Exhibit

Feb 2025 - Apr 2025

- Collaborated with a team of five to develop a museum exhibit for kids on sustainable energy sources for cities
- Utilized SolidWorks to design and fabricate hundreds of parts using a 3D printer and a laser cutter
- Coded and integrated a GUI using MATLAB and Arduino IDE to turn learning into a gamified experience

NASA Bridge (Cornerstone of Engineering)

Jan 2025 - Feb 2025

- Collaborated with a team of 4 to design a robust bridge for use on Mars with a 262:1 strength to weight ratio
- Used Fusion 360 to design and perform stress analysis on 7 different variations of the bridge